

Interview Problems

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 - a. Atmost 1's replace
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Notes



$$A[i] = \{0, 1\}$$

< Question > : Given a binary array $[]$. We can at most replace a single 0 with 1. Find the maximum consecutive 1's we can get in the array $[]$ after the replacement.

$$1 \leq N \leq 10^3$$

$A = [1, 1, 0, 1, 1, 0, 1, 1]$

 (Diagram: A horizontal array of 8 elements. The 3rd and 6th elements are 0. Above the 3rd element is a bracket labeled '1'. Above the 6th element is a bracket labeled '1'. Below the 3rd element is a bracket labeled '5'. Above the 4th to 8th elements is a bracket labeled '5'. To the right of the array is 'Ans = 5'.

Example : $[1, 1, 0, 1, 1, 0, 1, 1, 1]$

 (Diagram: A horizontal array of 9 elements. The 3rd and 6th elements are 0. Above the 3rd element is a bracket labeled '2'. Above the 6th element is a bracket labeled '6'. Below the 3rd element is a bracket labeled '5'. Below the 6th element is a bracket labeled '3'. To the right of the array is 'Ans = 6'.

Example : $[0, 1, 1, 1, 0, 1, 1, 0, 1, 1, 0]$

 (Diagram: A horizontal array of 11 elements. The 1st, 5th, 8th, and 11th elements are 0. Above the 5th element is a bracket labeled '5'. Below the 1st element is a bracket labeled '4'. Below the 2nd to 7th elements is a bracket labeled '6'. Below the 8th to 10th elements is a bracket labeled '3'. To the right of the array is 'Ans = 6'.

Example : $[1, 1, 1, 1, 1]$

 (Diagram: A horizontal array of 5 elements, all 1s. Below the entire array is a bracket labeled '5'. To the right of the array is 'Ans = 5'.

Example : $[0, 0, 0, 0, 0]$

 (Diagram: A horizontal array of 5 elements, all 0s. Above the 1st element is a bracket labeled '1'. To the right of the array is 'Ans = 1'.



0 1 2 3 4 5 6 7 8 9 10
 [0 1 1 1 0 1 1 0 1 1 0]
 ← 1 → ← 1 →

Bruteforce →

cnt1 = 0

for i → 0 to (N-1) {

 cnt1 += A[i]

}

TC = O(N)

if (cnt1 == N) return N

ans = 1

for i → 0 to (N-1) {

 if (A[i] == 0) {

count #1's
on left = l

 l = 0 j = i - 1

 while (j >= 0 && A[j] == 1) {

 l++ j--

 }

TC = O(N)

 r = 0 k = i + 1

 while (k < N && A[k] == 1) {

 r++ k++

 }

TC = O(N)

count #1's
on right = r



$$ans = \max(ans, l + 1 + r)$$

}
↑
}

$$TC = \cancel{O(N^2)} \rightarrow \underline{O(N)}$$

[0 1 1 1 0 1 1 0 1 1 0]



$$TC = O(3N) = \underline{O(N)}$$

$$SC = \underline{O(1)}$$

x x x x x x x x x x x x x

[1 1 0 0 1 1 1 0 1 0 0 0 1 1]





< Question > : Given a binary array $[]$. We can swap a single 0 with 1. Find the maximum consecutive 1's we can get in the array $[]$ after at most 1 swap.

$$A = [\cancel{1} 0 1 1 \cancel{0} 1 1] \quad \text{Ans} = \underline{5}$$

$\underset{0}{\phantom{\cancel{1}}}$ $\underset{1}{\phantom{\cancel{0}}}$

$$A = [0 0 \cancel{1} 1 \cancel{0} 1 1] \quad \text{Ans} = \underline{4}$$

$\underset{0}{\phantom{\cancel{1}}}$ $\underset{1}{\phantom{\cancel{0}}}$

$$A = [1 1 \cancel{0} 1 1 \cancel{1}] \quad \text{Ans} = \underline{5}$$

$\underset{1}{\phantom{\cancel{0}}}$ $\underset{0}{\phantom{\cancel{1}}}$

$$\begin{array}{cccccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \text{arr}[] \rightarrow & [& 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 &] \end{array} \rightarrow \begin{array}{l} \text{prev ans} = 6 \\ \text{cur ans} = 5 \end{array}$$

$$A = [1 0 1 1 0 1 1] \rightarrow \begin{array}{l} \text{prev ans} = 5 \\ \text{cur ans} = 5 \end{array}$$

```

cnt1 = 0
for i → 0 to (N-1) {
    cnt1 += A[i]
}

if (cnt1 == N) return N

```



ans = 0

for $i \rightarrow 0$ to $(N-1)$ {

if ($A[i] == 0$) {



l = 0 j = i - 1

while ($j \geq 0$ && $A[j] == 1$) {

l++ j--

}

r = 0 k = i + 1

while ($k < N$ && $A[k] == 1$) {

r++ k++

}

c = l + 1 + r

if ($c > \text{cnt} - 1$) c--

ans = max(ans, c)



}

}

TC = $O(N)$

SC = $O(1)$



Majority Element



< **Question** > : Given array [N]. Find the majority element



Elements which occurs more than $N/2$ times.

arr[] $\rightarrow$ [1, 2, 4] $\longrightarrow$ No majority element

arr[] $\rightarrow$ [3, 4, 3, 2, 4, 4, 4] Ans = 4 freq(4) > $N/2$

arr[] $\rightarrow$ [3, 3, 4, 2, 4, 4, 2, 4] $\longrightarrow$ No majority element

arr[] $\rightarrow$ [3, 4, 3, 6, 1, 3, 2, 5, 3, 3, 3] Ans = 3

arr[] $\rightarrow$ [4, 6, 5, 3, 4, 5, 6, 4, 4, 4] $\longrightarrow$ No majority element

$$\text{freq}(4) = 5 \neq \frac{N}{2} = 5$$



Observations :

At max, how many majority elements are possible?

Ans = 1



$$\text{freq}(\text{majority}) > N/2$$

Bruteforce $\rightarrow \forall A[i]$, count $\text{freq}(A[i])$ & check if it is $> N/2$.

a) Use hashmap to store frequency.

$$TC = \underline{O(N)}$$

$$SC = \underline{O(N)}$$

b) Sort & calculate freq. ✓

$$TC = \underline{O(N \log(N))}$$

$$SC = \underline{O(1)}$$

$A = [1 \quad 2 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 4]$

Ans = 3

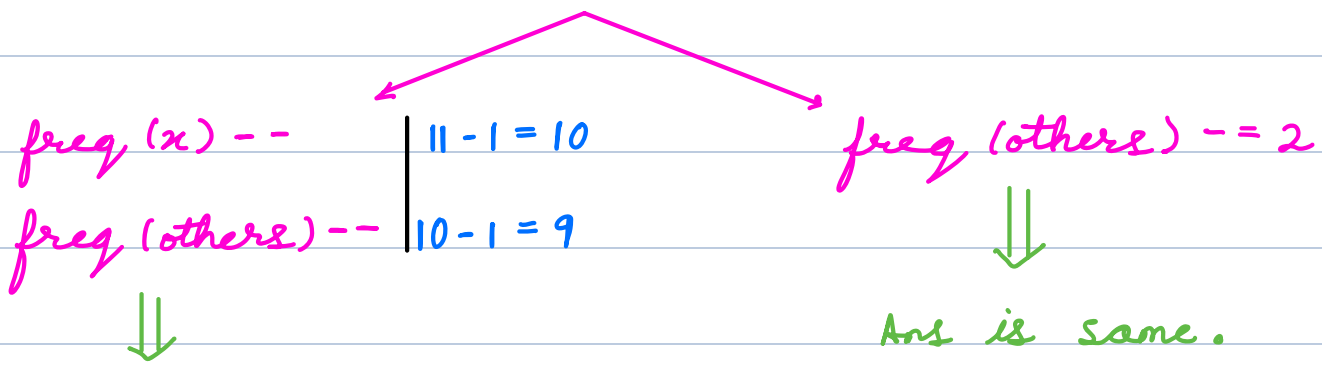
There can be only 1 majority element (let say 'x').

$$\text{freq}(x) > \frac{N}{2}$$

$$\text{freq}(\text{others}) < N/2$$



Remove any 2 distinct from the array.



Moore's Voting Algorithm

Remove two distinct elements till it is possible → Remaining element is in majority.

A = [3 4 3 3 5 6 4]

ans = 3 3 4

f = +0+2+0+1

verify if freq > N/2.

A = [3 4 3 3 3 6 4]

ans = 3 3 → freq(3) = 4 > 7/2 Ans = 3

f = +0+2+3+1

```

ans = -1    f = 0
for i → 0 to (N-1) {
    if (f == 0) {    ans = A[i]
                    f = 1    }
    else if (A[i] == ans)    f++
    else    f--
}

```

```

f = 0
for i → 0 to (N-1) {
    if (A[i] == ans)    f++
}

```

```

if (f > N/2)    return ans
else    return -1

```

TC = $O(N)$ SC = $O(1)$



< Question > : Given array [N] [M].

Make all elements in a row and column zero if $\text{arr}[i][j] = 0$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 0 \\ 9 & 2 & 0 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 0 \\ 9 & 2 & 13 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 9 & 2 & 13 & 0 \end{bmatrix}$$

$$\begin{array}{c} 0 \quad 1 \quad 2 \quad 3 \\ \begin{array}{c} 0 \\ 1 \\ 2 \end{array} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 0 \\ 9 & 2 & 13 & 4 \end{bmatrix} \end{array}$$

$$SC = O(1)$$

Sol $\rightarrow$ First convert all row/col to -1 (or ∞) except where element is 0,

$$\begin{array}{c} 0 \quad 1 \quad 2 \quad 3 \\ \begin{array}{c} 0 \\ 1 \\ 2 \end{array} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 0 \\ 9 & 2 & 13 & 4 \end{bmatrix} \end{array}$$

then update all -1's to 0.

$$TC = O(N^3) \quad [N \cong M]$$

$$SC = O(1)$$



$$\begin{array}{c} 0 \quad 1 \quad 2 \quad 3 \\ 0 \left[\begin{array}{cccc} -1 & 0 & -1 & 0 \\ 1 & 0 & 3 & 0 \\ 5 & 6 & 7 & 0 \\ 9 & 2 & 13 & 4 \end{array} \right] \\ 1 \\ 2 \end{array}$$

$$\begin{array}{c} 0 \quad 1 \quad 2 \quad 3 \\ 0 \left[\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 9 & 0 & 13 & 0 \end{array} \right] \\ 1 \\ 2 \end{array} \quad (Ans)$$
